Embedded System Design Parts Kit Contents Fall 2026
8/24/2026

Part Description	Quantity		Unit Price	Amount	Suggested Use	Location
Printed Circuit Board	1		$33.00	$33.00	Base Board (standard)	Big bag
16x4 LCD with HD44780 Compatible Controller	1		$8.00	$8.00	User Display	Big bag
40-Pin Wire Wrap DIP Socket (0.6" wide)	1		$3.00	$3.00	Microcontroller	Small bag 1
28-Pin Wire Wrap DIP Socket (0.6" wide)	1		$2.00	$2.00	NVRAM or EPROM and SRAM	Small bag 1
20-Pin Wire Wrap DIP Socket (0.3" wide)	4		$2.00	$8.00	74245, 74373, 74374, SPLD, Pull-ups	Small bag 1
16-Pin Wire Wrap DIP Socket (0.3" wide)	3		$1.50	$4.50	MAX232, I2C Expander, Decoding Logic	Small bag 1
14-Pin Wire Wrap DIP Socket (0.3" wide)	2		$1.00	$2.00	Discrete logic and components	Small bag 1
8-Pin Wire Wrap DIP Socket (0.3" wide)	1		$1.00	$1.00	Serial EEPROM	Small bag 1
Pin Strip Socket, Single Row, 14 Contacts	1		$2.00	$2.00	Solder into LCD to mate with pin strip header	Small bag 1
Pin Strip Header, Single Row, 36 Contacts	2		$2.50	$5.00	14 for LCD, 8/9 for RS-232, others for jumpers	Small bag 1
Aluminum Standoff, Female/Female, 6-32	4		$0.50	$2.00	Board Support (top)	Small bag 1
Aluminum Standoff, Male/Female, 6-32	4		$0.50	$2.00	Board Support (bottom)	Small bag 1
DE-9 Serial Connector	1		$1.00	$1.00	RS-232	Small bag 1
TO-220 Heat Sink	1		$0.50	$0.50	7805T +5V Regulator (optional)	Small bag 1
2.1mm Male Power Connector	1		$1.50	$1.50	Power Jack	Small bag 1
Toggle Switch, SPST or SPDT	1		$1.50	$1.50	Master Power On/Off Switch	Small bag 1
T44 Wire Wrap Pins	20		$0.10	$2.00	Discrete Component Mounting	Small bag 2
Pushbutton Switch, Momentary	4		$0.25	$1.00	Reset Circuit, Digital Input, FLIP Programming	Small bag 2
Jumper/Shorting Block/Shunt	5		$0.10	$0.50	8051 processor Flash programming, etc.	Small bag 2
7805T +5V Regulator, TO-220 Package	1		$0.25	$0.25	Power Supply	Small bag 2
11.0592 MHz Crystal	1		$1.00	$1.00	Microcontroller Clock Source	Small bag 2
27 pF Capacitor	3		$0.05	$0.15	Oscillator Circuit	Small bag 2
47 uF Electrolytic Capacitor	2		$0.10	$0.20	Power Supply Decoupling	Small bag 2
10 uF Electrolytic Capacitor	2		$0.10	$0.20	Reset Circuit, Power Supply Decoupling	Small bag 2
0.1 uF Monolithic Ceramic Capacitor (>=10V)	10		$0.10	$1.00	Board Decoupling	Small bag 2
1.0 uF Monolithic Ceramic Capacitor (>=16V)	5		$0.20	$1.00	MAX232 Charge Pump (5x1.0uF)	Small bag 2
4.7 uF Monolithic Ceramic Capacitor (>=10V)	2		$0.40	$0.80	8051 Processor Decoupling (SMT is an option)	Small bag 2
Diode, 1N4003 (1 Amp)	5		$0.10	$0.50	Voltage Regulator Protection	Small bag 2
Diode, 1N914 or 1N4148	1		$0.10	$0.10	Reset Circuit	Small bag 2
10K 1/4W Resistor	1		$0.10	$0.10	Reset Circuit	Small bag 2
4.7K (or 2.7K) Pull-up Resistors, 10-pin SIP	1		$0.25	$0.25	80C51 Port 0 Pull-ups, Other Pull-ups	Small bag 2
330 Ohm 1/4W Resistor	5		$0.10	$0.50	Current Limiting (LEDs, reset switch)	Small bag 2
5.1K Ohm 1/4W or 1/8W Resistor	2		$0.10	$0.20	EEPROM Control Line Pull-ups	Small bag 2
Potentiometer	1		$1.00	$1.00	LCD Contrast Control	Small bag 2
LED, Red	2		$0.10	$0.20	General Status Indicator	Small bag 2
LED, Green	2		$0.10	$0.20	General Status Indicator, Power On	Small bag 2
2N2222 NPN Transistor, TO-92	2		$0.25	$0.50	LED Driver, General Purpose	Small bag 2
2N2907 PNP Transistor, TO-92	2		$0.25	$0.50	LED Driver, General Purpose	Small bag 2
74LS373, Transparent Latch, Tri-state, DIP-20	1		$1.00	$1.00	Address Latch	Narrow tube
74LS374, Octal D Flip-Flop, Tri-state, DIP-20	1		$1.00	$1.00	General Purpose, LED Driver, Debug Status	Narrow tube
MAX 232, RS-232 Line Driver/Receiver, DIP-16	1		$1.50	$1.50	RS-232	Narrow tube
24C16 Serial EEPROM, DIP-8	1		$1.00	$1.00	Non-volatile Data Storage via I2C bus	Narrow tube
PCF8574 I2C I/O Expander, DIP-16	1		$2.00	$2.00	GPIO Expansion via I2C bus	Narrow tube
SPI DAC (e.g. Microchip MCP4802), DIP-8	1		$2.00	$2.00	Data Conversion via SPI bus	Narrow tube
SPLD (e.g. Microchip ATF16V8B/C), DIP-20	1		$2.00	$2.00	Programmable Logic, glue logic	Narrow tube
8051, (Microchip/Atmel) AT89C51RC2, DIP-40	1		$10.00	$10.00	Microcontroller	Wide tube
32KB NVRAM, DIP-28	0		$30.00	$0.00	Variable/Data Storage	Tool kit (sign out)
32KB SRAM, DIP-28	0		$10.00	$0.00	Data/Code Storage	Tool kit (sign out)
28-Pin ZIF (Zero Insertion Force) Socket	0		$15.00	$0.00	NVRAM/EPROM	Tool kit (sign out)
Wire Wrap Wire (30AWG Kynar, 1 foot unit)	100		$0.10	$10.00	Signal and Power Connections	1B28 Lab

Total Item Value **$119.65** **Note: Student Cost for Parts Kit is $100.00**

Students may sign out an accessory/tool kit containing the following items:
NVRAM, 28-pin ZIF socket (for NVRAM/EPROM), wire wrap tool, power supply, cutters, pliers, serial cable, USB-RS232 converter cable

ARM Dev Kit (e.g. STM32) + USB cable	$30.00	$0.00	ARM lab assignments
Silabs 8051 Dev Kit (e.g. EFM8BB1) + USB microB cable	$25.00	$0.00	8051 lab assignments

Portable logic analyzers are also available to sign out for the semester. Items must be returned by the end of the semester.
Other items available to sign out or use may include: digital logic probe, higher frequency oscope probes, hot air rework station
Students are financially responsible for any items they sign out.
Resistors, caps, wire wrap wire, and some small parts are available in the laboratory.
Note: There may be a few part substitutions or omissions, depending on availability.
Students should plan on obtaining parts for the final project from other sources, such as the ECEE Store, Digi-Key, Mouser, etc.